

Study Report: Power BI Analyst - Career Path & Core Skills

Written by Gagan Pasupuleti

Family:	Power BI Analyst
Level:	Beginner to Advanced
Tracks:	Beginner, Intermediate, Advanced
Chapters:	18
Source:	Official docs, free libraries, and industry-standard references (see sources and books)

Official sources and free libraries

- <https://learn.microsoft.com/en-us/power-bi/>
- <https://learn.microsoft.com/en-us/power-query/>
- <https://dax.guide/>
- <https://learn.microsoft.com/en-us/fabric/>
- <https://www.sqlbi.com/articles/>
- <https://learn.microsoft.com/en-us/power-bi/guidance/>

Summary

This report covers the Power BI Analyst role with Power BI Desktop, Power Query (M), DAX, dimensional modeling, and Microsoft Fabric fundamentals (2024+). Topics include semantic models, time intelligence, incremental refresh, lakehouse integration, deployment pipelines, and enterprise governance for self-serve analytics.

Skill stack by level

Beginner

- Power BI Desktop interface
- Import vs DirectQuery modes
- Power Query basic transforms
- Star schema modeling
- Basic DAX measures (SUM, DIVIDE)
- Chart and slicer design
- Publish to Power BI Service

Intermediate

- Advanced Power Query (M) patterns
- DAX CALCULATE and time intelligence
- Row-level security (RLS)
- Incremental refresh policies

- Microsoft Fabric Lakehouse (2024+)
- Deployment pipelines (dev/test/prod)
- Performance Analyzer and DAX Studio
- XMLA endpoint connectivity

Advanced

- Composite models and aggregation tables
- Fabric Data Factory pipelines
- Real-Time Intelligence (KQL, 2024+)
- Power BI Embedded and REST APIs
- Purview lineage and governance
- Premium capacity sizing
- CI/CD with Tabular Editor 3
- Enterprise certification workflows

Recommended books

1. Beginning Power BI - Mark Pagani

Level: Beginner | Power BI Desktop basics: import, model, visualize, publish.

2. The Definitive Guide to DAX - Marco Russo & Alberto Ferrari

Level: Intermediate | DAX measures, CALCULATE, time intelligence, and filter context.

3. M Is for (Data) Monkey - Ken Puls & Miguel Escobar

Level: Intermediate | Power Query M language for data transformation.

4. Microsoft Fabric Fundamentals - Microsoft Learn

Level: Intermediate | Unified analytics: lakehouse, pipelines, and Power BI in Fabric.

5. Analyzing Data with Power BI and Power Pivot - Marco Russo & Alberto Ferrari

Level: Advanced | Data modeling and DAX for complex enterprise models.

End-to-end projects

Project 1: Retail Sales Dashboard

Beginner | 2 weeks - Power BI report from Excel/CSV: Power Query cleanup, star schema, DAX measures, published to Service.

- Import: Clean query
- Model: Validated model
- DAX: Measure list doc
- Publish: Live dashboard URL

Project 2: Multi-Source Finance Model

Intermediate | 3-4 weeks - Merge SQL Server + Excel + API data; complex DAX time intelligence; row-level security by department.

- ETL: Combined dataset
- RLS: RLS test matrix
- Time Intel: Finance KPI page
- Tune: Before/after perf screenshot

Project 3: Microsoft Fabric Lakehouse Analytics

Advanced | 5-6 weeks - Build Fabric lakehouse: ingest raw data, transform in notebooks, semantic model, and executive dashboard.

- Lakehouse: Lakehouse schema
- Pipeline: Pipeline run history
- Semantic: Semantic model
- Dashboard: Published Fabric report

Chapters

Track: Beginner

Chapter 1: Power BI Analyst Role Overview [Beginner]

Chapter focus: Power BI Analyst Role Overview *(Level: Beginner)**

Power BI analysts bridge data engineering and business users. You model data, write DAX, design reports, and publish to the Power BI Service. With Microsoft Fabric (2024+), the role extends into lakehouse ingestion - still centered on trusted semantic models.

Code Reference:

```
-- Power BI connects to hundreds of sources
-- Typical flow: Get Data -> Transform -> Model -> Report -> Publish
```

What it shows: The analyst workflow is linear but iterative - model changes ripple to all reports.

Topic guide

What it actually is

A Power BI analyst builds self-serve analytics on Microsoft's BI stack.

When to use it

Organizations standardized on Microsoft 365, Azure, and Fabric.

Where to use it

Finance reporting, sales ops, HR analytics, and operational KPIs.

Real use example

A regional manager filters a published report by territory without emailing IT for a CSV.

Chapter 2: Connecting to Data Sources [Beginner]

Chapter focus: Connecting to Data Sources *(Level: Beginner)**

Power BI connects to Excel, SQL Server, Snowflake, SharePoint, and Fabric OneLake. Choose Import for speed, DirectQuery for real-time, or Direct Lake (Fabric) for large Delta tables. Document gateway requirements for on-prem sources.

Code Reference:

```
let
    Source = Sql.Database("server", "dw"),
    Sales = Source{[Schema="dbo", Item="fact_sales"]}[Data]
in
    Sales
```

What it shows: Power Query M loads SQL table fact_sales - first step in most enterprise models.

Topic guide**What it actually is**

Get Data establishes the pipeline from source systems into Power BI.

When to use it

Every new dashboard starts with source connections.

Where to use it

SQL warehouses, Excel finance files, and OData APIs.

Real use example

Analyst connects Snowflake marketing spend to internal CRM for unified campaign ROI.

Chapter 3: Power Query Transformations [Beginner]**Chapter focus: Power Query Transformations** *(Level: Beginner)**

Power Query is click-first ETL: remove columns, filter rows, split columns, unpivot. Applied Steps are reproducible - rename steps clearly. Prefer Table.Buffer for small lookups, not million-row tables.

Code Reference:

```
let
    Source = Csv.Document(File.Contents("orders.csv")),
    Promoted = Table.PromoteHeaders(Source),
    Typed = Table.TransformColumnTypes(Promoted, {{"amount", type number}})
in
    Typed
```

What it shows: PromoteHeaders + type coercion prevents DAX treating amounts as text.

Topic guide**What it actually is**

Power Query cleans and shapes data before modeling.

When to use it

When source exports need cleansing every refresh.

Where to use it

Weekly CSV drops, API JSON flattening, and Excel consolidation.

Real use example

Unpivoting wide month columns turns twelve columns into a tidy date/value pair for charting.

Chapter 4: Star Schema Data Modeling [Beginner]

Chapter focus: Star Schema Data Modeling** *(Level: Beginner)

Star schemas place metrics in fact tables and descriptors in dimensions. Use surrogate keys, conformed date tables, and single-direction relationships. Avoid snowflaking unless saving meaningful memory - simplicity aids DAX.

Code Reference:

```
# Model checklist
# - One date table marked as date table
# - Facts at transaction grain
# - No bi-directional unless expert reason
```

What it shows: A checklist prevents relationship mistakes that inflate totals.

Topic guide

What it actually is

Dimensional modeling organizes data for fast filtering and correct aggregation.

When to use it

Import and Direct Lake semantic models in Power BI.

Where to use it

Retail sales, subscription billing, and inventory snapshots.

Real use example

A single fact_orders connects to dim_customer and dim_date - slicers filter revenue correctly.

Chapter 5: Report Design and UX Basics [Beginner]

Chapter focus: Report Design and UX Basics** *(Level: Beginner)

Use consistent fonts, 8-12 visuals per page max, and purposeful color. Slicers sync across pages via Sync Slicers pane. Bookmarks capture narrative states for guided storytelling in executive reviews.

Code Reference:

```
# Visual hierarchy
# 1. KPI card (big number)
# 2. Trend line (context)
# 3. Breakdown bar (driver)
```

What it shows: Hierarchy guides the eye - biggest number answers the headline question first.

Topic guide**What it actually is**

Report design translates the model into decisions.

When to use it

Every published Power BI report and paginated export.

Where to use it

Sales flash, pipeline reviews, and classroom analytics.

Real use example

Bookmark toggles 'by region' vs 'by product' without duplicating pages.

Chapter 6: Introduction to DAX Measures [Beginner]**Chapter focus: Introduction to DAX Measures** *(Level: Beginner)**

DAX calculates on the model, not cell-by-cell like Excel. Measures evaluate in filter context. Start with SUM, AVERAGE, DISTINCTCOUNT; always use DIVIDE for ratios to handle divide-by-zero.

Code Reference:

```
Total Sales = SUM ( Sales[Amount] )
Margin % = DIVIDE ( [Total Sales] - [Total Cost], [Total Sales] )
```

What it shows: DIVIDE returns blank instead of error when sales are zero.

Topic guide**What it actually is**

DAX is the formula language for Power BI and Analysis Services tabular models.

When to use it

Whenever Excel-style calculated columns are too heavy or inflexible.

Where to use it

KPI cards, matrix visuals, and time comparisons.

Real use example

Margin % measure works on any visual - one definition, consistent boardroom numbers.

Track: Intermediate

Chapter 7: Advanced Power Query (M) Patterns [Intermediate]**Chapter focus: Advanced Power Query (M) Patterns** *(Level: Intermediate)**

Parameterize file paths and server names for portability. Use List.Generate for pagination APIs; Table.NestedJoin for complex merges. Enable query folding - check View Native Query to push

work to SQL.

Code Reference:

```
let
    Page = (n as number) =>
        Json.Document(Web.Contents("https://api.example.com/items?page=" & Text.From(n))),
    Pages = List.Generate(() => [i=1, data=Page(1)], each [data] <> null, each [i=[i]+1, data=Page([i]+1)])
in
    Table.FromList(Pages, each [data][items])
```

What it shows: List.Generate loops API pages - common pattern for marketing APIs.

Topic guide

What it actually is

Advanced M handles APIs, parameters, and folding-aware transforms.

When to use it

Refreshing models from REST APIs or multi-file folders.

Where to use it

Marketing spend APIs, blob folders, and incremental file drops.

Real use example

Parameterized server name lets dev/test/prod switch without rewriting queries.

Chapter 8: DAX CALCULATE and Filter Context [Intermediate]

Chapter focus: DAX CALCULATE and Filter Context** *(Level: Intermediate)

CALCULATE modifies filter context - the heart of DAX. Sales YTD = TOTALYTD ([Total Sales], 'Date'[Date]). Understand row context vs filter context before iterator functions (SUMX).

Code Reference:

```
Sales CY = CALCULATE ( [Total Sales], 'Date'[Year] = 2025 )
Sales PY = CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( 'Date'[Date] ) )
YoY % = DIVIDE ( [Sales CY] - [Sales PY], [Sales PY] )
```

What it shows: SAMEPERIODLASTYEAR shifts the date filter for year-over-year comparisons.

Topic guide

What it actually is

CALCULATE applies filters to measures dynamically.

When to use it

Time intelligence, conditional KPIs, and % of total metrics.

Where to use it

Finance close packs, retail comps, and subscription growth boards.

Real use example

YoY % measure works on any visual slice - region, product, or rep.

Chapter 9: Row-Level Security (RLS) [Intermediate]**Chapter focus: Row-Level Security (RLS)** *(Level: Intermediate)**

RLS restricts data rows per audience. Define roles with DAX filters on tables: [Email] = USERPRINCIPALNAME(). Test with View as Role before publish. Dynamic RLS via security bridge tables scales to thousands of users.

Code Reference:

```
[Region] = LOOKUPVALUE ( Users[Region], Users[Email], USERPRINCIPALNAME() )
```

What it shows: LOOKUPVALUE maps the logged-in user to their allowed region.

Topic guide**What it actually is**

RLS enforces data access inside the semantic model.

When to use it

Multi-tenant SaaS analytics and regional sales reports.

Where to use it

Franchise dashboards, partner portals, and HR sensitive metrics.

Real use example

A store manager sees only their district - same report, different rows.

Chapter 10: Incremental Refresh and Performance [Intermediate]**Chapter focus: Incremental Refresh and Performance** *(Level: Intermediate)**

Incremental refresh loads only new/changed partitions - essential for large facts. Requires date column, RangeStart/RangeEnd parameters, and Premium capacity. Use Performance Analyzer and DAX Studio to find slow visuals.

Code Reference:

```
let
    Filtered = Table.SelectRows(Source, each [OrderDate] >= RangeStart and [OrderDate] < RangeEnd)
in
    Filtered
```

What it shows: Range parameters align Power Query filter with incremental policy windows.

Topic guide**What it actually is**

Incremental refresh keeps large models fast and refresh windows small.

When to use it

Fact tables with millions+ rows and daily append patterns.

Where to use it

E-commerce orders, clickstream aggregates, and IoT telemetry.

Real use example

Daily refresh drops from 2 hours to 12 minutes after partitioning by order_date.

Chapter 11: Microsoft Fabric Lakehouse Basics (2024+) [Intermediate]**Chapter focus: Microsoft Fabric Lakehouse Basics (2024+)** *(Level: Intermediate)**

Fabric unifies data engineering and BI on OneLake. A Lakehouse stores Delta/Parquet files with SQL analytics endpoint. Shortcuts attach ADLS/S3 without copying data. Power BI Direct Lake queries Delta files in place - near Import speed.

Code Reference:

```
-- Fabric SQL endpoint
SELECT product_id, SUM(quantity) AS units
FROM silver.fact_orders
GROUP BY product_id;
```

What it shows: SQL endpoint lets analysts validate silver tables before modeling.

Topic guide**What it actually is**

Fabric Lakehouse is Microsoft's open-format storage layer for analytics.

When to use it

Greenfield analytics platforms on Microsoft cloud (2024+).

Where to use it

Replacing fragmented ADF + AS + Power BI setups.

Real use example

Marketing uploads CSV to Files area; notebook promotes to silver; Power BI reads via Direct Lake.

Chapter 12: Fabric Real-Time Intelligence Preview [Intermediate]**Chapter focus: Fabric Real-Time Intelligence Preview** *(Level: Intermediate)**

Fabric Real-Time Intelligence (2024-2025) adds Eventstreams and KQL Database for streaming logs. Power BI can query hot cache for operational dashboards. Pair with Activator for threshold alerts.

Code Reference:

```
.create table pageviews (timestamp: datetime, user_id: string, path: string)
```

What it shows: KQL table schema defines streaming events landing from Eventstream.

Topic guide

What it actually is

Real-Time Intelligence handles sub-minute latency analytics in Fabric.

When to use it

Ops monitoring, fraud signals, and live campaign tracking.

Where to use it

Web telemetry, manufacturing sensors, and support queue depth.

Real use example

Activator emails on-call when error rate > 2% in rolling 15 minutes.

Chapter 13: Deployment Pipelines and ALM [Intermediate]

Chapter focus: Deployment Pipelines and ALM** *(Level: Intermediate)

Deployment pipelines promote content dev ? test ? prod with diff review. Pair with Git integration (2024+) for source control. Tabular Editor scripts automate measure renames across environments.

Code Reference:

```
# Promotion checklist
# 1. Run BPA rules in Tabular Editor
# 2. Compare schema diff
# 3. Deploy with notes
```

What it shows: Best Practice Analyzer catches modeling issues before production.

Topic guide

What it actually is

ALM governs how reports and datasets move across environments.

When to use it

Any team with multiple analysts and production subscribers.

Where to use it

Enterprise BI centers of excellence.

Real use example

A breaking relationship change is caught in test - prod subscribers never see blank KPIs.

Track: Advanced

Chapter 14: Composite Models and Aggregations [Advanced]

Chapter focus: Composite Models and Aggregations *(Level: Advanced)**

Composite models mix Import and DirectQuery tables. Aggregation tables pre-summarize facts for drill-down performance. User hierarchies hit agg first, then detail - transparent to report users.

Code Reference:

```
Sales Agg = SUMMARIZE ( Sales, Date[Month], Product[Category], "Amount", SUM ( Sales[Amount] ) )
```

What it shows: SUMMARIZE builds an aggregation table DAX engine can substitute automatically.

Topic guide**What it actually is**

Composite + aggregation optimizes large models without losing detail.

When to use it

When Import models exceed memory or DirectQuery is too slow for drill-down.

Where to use it

Global retail chains and multi-year telemetry models.

Real use example

Retail BI team uses agg table for monthly scans - detail loads only on drill-through.

Chapter 15: Advanced DAX: Context Transition [Advanced]**Chapter focus: Advanced DAX: Context Transition** *(Level: Advanced)**

Calculated columns evaluate row-by-row; measures use filter context. Context transition occurs when CALCULATE iterates - a common source of bugs. Use variables (VAR) for readability and single evaluation.

Code Reference:

```
Rank in Category =  
VAR CatSales = CALCULATE ( [Total Sales], ALLEXCEPT ( Product, Product[Category] ) )  
RETURN RANKX ( ALL ( Product[Product] ), CatSales, , DESC, Dense )
```

What it shows: VAR stores intermediate results - easier to debug than nested CALCULATE.

Topic guide**What it actually is**

Advanced DAX handles ranking, moving averages, and dynamic segmentation.

When to use it

When CALCULATE-heavy measures misbehave or rankings are needed in visuals.

Where to use it

Subscription analytics with tiered discount logic.

Real use example

VAR-based rank measure finally matches Excel prototype after context transition fix.

Chapter 16: Fabric Data Factory Pipelines [Advanced]**Chapter focus: Fabric Data Factory Pipelines** *(Level: Advanced)**

Fabric Data Factory (successor patterns to ADF in Fabric) orchestrates copy, notebook, and Dataflow Gen2 activities on OneLake. Schedule pipelines for bronze ingestion before semantic refresh.

Code Reference:

```
# Pipeline: Copy blob -> Notebook cleanse -> Refresh dataset
# Trigger: daily 05:00 UTC
```

What it shows: Orchestration ensures silver tables update before Power BI refresh fires.

Topic guide**What it actually is**

Data Factory pipelines automate lakehouse ingestion in Fabric.

When to use it

When bronze/silver loads must run before semantic model refresh on OneLake.

Where to use it

Nightly ERP extracts and multi-source consolidation.

Real use example

Pipeline failure blocks dataset refresh - ops gets alert before executives see stale KPIs.

Chapter 17: Power BI Embedded and REST API [Advanced]**Chapter focus: Power BI Embedded and REST API** *(Level: Advanced)**

Embed reports in customer-facing apps with app-owns-data or user-owns-data models. REST API automates workspace management, refresh triggers, and export to PDF. Service principals require tenant admin approval - document security review.

Code Reference:

```
POST https://api.powerbi.com/v1.0/myorg/groups/{id}/datasets/{id}/refreshes
Authorization: Bearer {token}
```

What it shows: Refresh API kicks dataset update after upstream pipeline completes.

Topic guide**What it actually is**

Embedded analytics puts Power BI inside SaaS products.

When to use it

When customers need in-app dashboards without separate Power BI licenses per user.

Where to use it

B2B SaaS dashboards, partner extranets, and mobile app WebViews.

Real use example

SaaS embeds revenue dashboard - tenant isolation enforced via RLS embed token.

Chapter 18: Enterprise Governance with Microsoft Purview [Advanced]

Chapter focus: Enterprise Governance with Microsoft Purview *(Level: Advanced)**

Purview scans Fabric items for lineage and sensitivity labels. Endorse datasets as Certified after BPA pass and owner sign-off. Monitor Premium capacity via Fabric Capacity Metrics app (2024+).

Code Reference:

```
# Certified dataset requirements
# - Owner assigned
# - Description on all tables
# - RLS tested
# - Refresh SLA documented
```

What it shows: Certification signals trust - analysts stop rebuilding the same model.

Topic guide

What it actually is

Governance ensures trustworthy, discoverable BI at scale.

When to use it

When workspace sprawl creates duplicate datasets and compliance questions.

Where to use it

Regulated industries and global rollouts.

Real use example

Purview lineage proves revenue metric traces to audited Fabric lakehouse table.